Quantum Circuit Optimization Using Geometric Programming: Convex Duration Allocation on Fixed NISQ Circuits

Youssef Ahmed Ramadan

Department of Artificial Intelligence Engineering, School of Engineering
University of Science and Technology at Zewail City
Giza, Egypt
Email: p-youssef.ramdan@zewailcity.edu.eg

***Abstract*— We study gate-duration allocation on a fixed quantum-circuit dependency graph under a declared inverse-duration plus exposure-error model. The continuous problem admits geometric-program and ordinary convex formulations; a path-flow dual bounds its unconstrained variant. For Clifford circuits with independent isotropic Pauli channels, backward Pauli tracking yields exact convex stabilizer-expectation constraints within the model, and a geometric-grid rule preserves feasible constraints under a sufficient safety margin. In a saved IBM Heron calibration-snapshot study, a model-attainable 24-instance protocol produced feasible solves through 128 qubits. Median solve times at 32, 64, and 128 qubits were 1.17, 2.95, and 6.41 s; median modeled makespan reductions against an optimal uniform-duration policy were 17.3%, 24.5%, and 24.5% under identical targets. A held-out fit to published measured single-qubit duration sweeps exhibits substantial residuals relative to reported uncertainties. Matched local simulations compare idealized GP choices with Qiskit ASAP/ALAP and dynamical decoupling; a ranking reversal under shot-static detuning limits generality. A first prespecified fixed-native-gate QPU check misses its 0.02 probability tolerance in all four families. After a backend calibration update, a separate six-family check meets that descriptive tolerance, but direct readout recalibration worsens mean absolute error relative to the new snapshot. The contradictory checks show sensitivity to calibration and do not identify the inverse-duration law. A training-selected, held-out test of two native X implementations on one physical qubit found that a 1,000-ns X gate improved X-basis return probability by 4.25 percentage points over equal-duration two-X dynamical decoupling at 48 logical X gates, with all three randomized blocks agreeing; its ranking reversed on Z-basis input. This is a physical gate-choice effect, not a validation of the GP duration policy, whose device-level benefit remains untested.**

***Index Terms*—** Quantum compilation, geometric programming, convex optimization, gate scheduling, Markovian noise, pulse libraries, Pauli tracking, Heron architecture.

I. Introduction

QUANTUM compilation connects abstract circuits to executable pulses and schedules. Gate count, depth, makespan, and noisy-state fidelity are distinct quantities. Shorter schedules can demand stronger control and may increase error. We therefore study a precise subproblem: duration allocation on a fixed dependency graph under a declared noise model. Its optimum and error bounds are conditional on that model, rather than guarantees about an uncharacterized device.

The mathematical structure of time-cost tradeoffs on directed acyclic dependency graphs (DAGs) traces back to the foundational network-flow duality of Kelley [1] and Fulkerson [2]. Geometric programming (GP) provides an established positive-variable optimization language with an exact convex transformation under logarithmic change of variables [3]. Its historical application to digital CMOS circuit sizing and timing closure revolutionized classical electronic design automation (EDA) [4]. In quantum computing, when gate durations are fixed, makespan minimization reduces to a longest-path computation on a DAG. However, physical quantum devices operate under a different decision space: continuous gate durations and discrete pulse implementations that preserve target local unitaries while trading off drive strength, environmental decoherence, and circuit latency.

Our contribution is a model-based timing method, rather than a claim of optimal circuit synthesis or hardware performance. First, we formulate fixed-DAG duration allocation as convex optimization and derive a path-flow lower bound. Second, under an explicitly post-gate isotropic Pauli channel model, we combine Clifford observable tracking with duration optimization and a conditional quantization guarantee. Third, we distinguish model-consistency checks from external validation, report matched-deadline comparisons, and supply an additional feasible large-circuit solve study. The inverse-duration coefficient is not inferred from the saved backend snapshot: its share is swept as an assumption. Claims of physical advantage therefore require measured duration–error curves and hardware-compatible controls.

Sections II–VII describe prior work, the mathematical model, its bounds, and the observable formulation. Sec-

tion VIII reports local simulations, including the feasible large-circuit study. Section IX describes the analysis pass and limitations; Section X concludes.

II. Related Work and Scientific Context

A. Duration-Aware Quantum Scheduling

Lussi, de Santiago, and Duzzioni recently introduced duration-aware quantum scheduling by assigning fast pulse implementations to critical-path operations and robust, longer implementations to available off-critical slack [5]. Our work builds upon this insight by formulating continuous duration allocation as an exact convex program, deriving analytical primal-dual optimality certificates, and incorporating nonunitary open-system channel bounds. In quantum compiler systems, noise-adaptive mapping and crosstalk-aware scheduling have been investigated by Murali et al. [12], [13], while pulse-efficient transpilation was developed by Earnest et al. [14].

B. Convex Scheduling and Classical Foundations

Convex project time-cost optimization on DAGs originates in the classic network-flow duality of Kelley [1] and Fulkerson [2], subsequently generalized to convex cost dual network flow by Ahuja et al. [6]. Geometric programming supplies an exact convex representation when all cost terms are positive monomials or posynomials [3]. For the proposed inverse-duration-plus-duration costs, the original variables also yield an ordinary convex formulation, establishing that efficiency stems from convexity rather than logarithmic change of variables alone.

C. Noise Modeling and Stabilizer Observables

The no-jump branch of a quantum-trajectory representation underlies the channel certificate [7]. Conditional-gradient methods provide established optimization-gap machinery [8]. Compositional robustness for quantum programs has been formalized by Hung et al. [10], while robust quantum control via sequential convex programming was investigated by Kosut et al. [11]. Sparse Pauli-Lindblad noise characterization has been established by van den Berg et al. [15]. Recently, P. Aigner et al. [16] developed analytical and compressed simulation techniques for noisy stabilizer circuits. Our framework leverages duration-dependent noise sensitivity to convert analytical stabilizer tracking into an *exact posynomial constraint*, enabling convex interior-point optimization for compiler scheduling.

III. Continuous Duration Model and Exact Convexity

Let a native circuit contain m unitary gates and a dependency DAG $\mathcal{D}$ with r edges. Per-qubit order is fixed by the logical circuit; concurrent operations on overlapping qubit supports are forbidden. Gate i has continuous duration $\tau_i \in [\ell_i, u_i]$, with $0 < \ell_i \le u_i$, generated by a Hermitian Hamiltonian $H_i = K_i/\tau_i$ such that

$$e^{-\mathrm{i}K_i} = e^{\mathrm{i}\phi_i} U_i, \quad K_i = \theta_i P_i/2, \tag{1}$$

where P_i denotes the generating Hamiltonian. On superconducting transmon processors such as IBM Heron, native operations comprise single-qubit rotations (SX, X, and virtual RZ) and two-qubit Controlled-Z (CZ) entangling interactions; entangling gates like RXX decompose into native CZ and single-qubit rotations. Because K_i is fixed, the ideal local unitary is duration-independent.

Define the integrated squared control effort $E(\tau) = \sum_i \kappa_i^2/\tau_i$, where $\kappa_i = \|K_i\|_2$. For a schedule with makespan M, the integrated noise hazard satisfies

$$\Lambda \le R(\tau, M) = bM + \sum_i \left(\frac{a_i}{\tau_i} + \gamma_i \tau_i \right), \tag{2}$$

where $b, a_i, \gamma_i > 0$. The inverse-plus-linear term $a_i/\tau_i + \gamma_i\tau_i$ represents a stylized phenomenological noise model capturing the fundamental engineering tension between drive-power-induced errors (leakage, non-adiabaticity, and amplifier distortion at small τ_i) and environmental decoherence exposure (T_1, T_2 relaxation and dephasing at large τ_i).

The objective balances noise hazard, makespan, and control effort:

$$J(\tau, M) = w_R \frac{R}{R_0} + w_M \frac{M}{M_0} + w_E \frac{E}{E_0}, \tag{3}$$

subject to precedence constraints $f_i + \tau_j \le f_j$ for $(i \to j) \in \mathcal{D}$, completion constraints $\tau_i \le f_i \le M$, and duration bounds $\ell_i \le \tau_i \le u_i$.

Theorem 1 (Convexity and Conditional Uniqueness)**.** *Suppose $R_0, M_0, E_0 > 0$, $w_R, w_M, w_E \ge 0$, and the gate coefficients are nonnegative. The fixed-DAG duration-allocation problem in* (3) *is an exact geometric program whenever the positive-variable formulation is used, and is convex in the original durations and finish-time variables. Define $A_i = w_R a_i/R_0 + w_E \kappa_i^2/E_0$. If $A_i > 0$ for every nonfixed duration coordinate, the optimal duration vector is unique. This does not imply unique intermediate finish times.*

Proof. Write $B = w_R b/R_0 + w_M/M_0 \ge 0$ and $C_i = w_R\gamma_i/R_0 \ge 0$. Eliminating the finish-time variables at fixed τ gives the reduced objective

$$F(\tau) = \sum_i (A_i/\tau_i + C_i\tau_i) + B \max_{p \in \mathcal{P}} p^\top \tau.$$

The box is compact, all reciprocal terms are convex on positive durations, and the maximum of path sums is convex. Hence a minimizer exists. If every free-coordinate $A_i > 0$, the Hessian of $\sum_i A_i/\tau_i$ is diagonal with entries $2A_i/\tau_i^3 > 0$ on the free-coordinate subspace. The reduced objective is then strictly convex, so its minimizing duration vector is unique. For the GP representation, positive monomials and posynomials express the objective and precedence inequalities; logarithmic substitution yields convex log-sum-exp inequalities. Fixed coordinates can be eliminated. A

zero coefficient or weight may remove strict convexity, so uniqueness is not asserted outside the stated condition. □

Proposition 1 (Positive Inverse-Power Extension)**.** *For a declared exponent $p > 0$, replace the reduced duration objective by*

$$F_p(\tau) = \sum_i \left(A_i \tau_i^{-p} + C_i \tau_i\right) + B \max_{q \in \mathcal{P}} q^\top \tau, \tag{4}$$

with $A_i, C_i \geq 0$, $B > 0$, and positive duration bounds. This is a geometric program and is convex in the original durations. If $A_i > 0$ for each free coordinate, the optimal duration vector is unique. The path-flow dual of Theorem 2 remains valid with coordinate response

$$\tau_i(\rho) = \operatorname{clip}\left[\left(\frac{pA_i}{C_i + \rho_i}\right)^{1/(p+1)}, \ell_i, u_i\right] \tag{5}$$

when $C_i + \rho_i > 0$; if it is zero and $A_i > 0$, the response is u_i.

Proof. Each $A_i \tau_i^{-p}$ is a positive monomial, hence the positive-variable objective and precedence ratios satisfy GP grammar. Its second derivative is $p(p+1)A_i \tau_i^{-p-2} \geq 0$, strictly positive on free coordinates with $A_i > 0$. The maximum of path sums is convex, so strict convexity and uniqueness follow as in Theorem 1. Minimax interchange over the compact duration box and path-flow polytope is unchanged. Differentiating $A_i/t^p + (C_i + \rho_i)t$ gives the displayed clipped minimizer. This extension changes the assumed error law; it does not validate any exponent from hardware data. □

Proposition 2 (Rectangular Box Robustness)**.** *If each coefficient in* (2) *belongs independently to a declared interval $[a_i^{\min}, a_i^{\max}]$, its maximum over that rectangle is attained at the upper endpoints. Replacing coefficients by those endpoints yields an exact robust GP.*

Proof. Each coefficient multiplies a nonnegative term, so R is coordinatewise monotone. The upper endpoint vector belongs to the uncertainty rectangle and attains its maximum for every feasible schedule. □

Proposition 3 (Finite-Scenario Robust GP)**.** *Let a declared finite set $\mathcal{S}$ contain nonnegative coefficient pairs (A_{is}, C_{is}) for a common exponent $p > 0$. Under the same positive duration box, DAG constraints, and optional deadline, the minimax problem*

$$\min_{\tau, M} \left\{ \max_{s \in \mathcal{S}} \sum_i (A_{is} \tau_i^{-p} + C_{is} \tau_i) + BM \right\} \tag{6}$$

is an exact geometric program in epigraph form. Its optimum is robust only with respect to the declared *scenarios; it is a physical error bound only if the true coefficients are contained in a justified uncertainty set.*

Proof. Introduce $z > 0$ and require $\sum_i (A_{is} \tau_i^{-p} + C_{is} \tau_i)/z \leq 1$ for each $s \in \mathcal{S}$. Every left-hand side is a posynomial, and $z + BM$ is a posynomial objective. The timing and deadline constraints retain GP form. For fixed feasible τ, the smallest feasible z is exactly the maximum scenario error, proving epigraph equivalence. Taking a maximum over finitely many convex scenario errors also proves convexity in the original durations. Nothing in this argument assigns statistical coverage to $\mathcal{S}$. □

Proposition 4 (Quantization Rounding Certificate)**.** *If a rounded vector τ' remains within $[\ell, u]$, then*

$$|R(\tau') - R(\tau)| \leq \sum_i \left(\frac{a_i}{\ell_i^2} + \gamma_i + b\right) |\tau_i' - \tau_i|. \tag{7}$$

Proof. Every path sum changes by at most $\sum_i |\tau_i' - \tau_i|$, bounding makespan change. The derivative magnitude of $a_i/\tau_i + \gamma_i \tau_i$ on $[\ell_i, u_i]$ is bounded by $a_i/\ell_i^2 + \gamma_i$. Summing yields the result. □

IV. Path-Flow Dual and Algorithm

Let $\mathcal{P}$ denote the set of all root-to-sink path incidence vectors in $\mathcal{D}$. Makespan is $M(\tau) = \max_{p \in \mathcal{P}} p^\top \tau$. Combining objective coefficients into $A_i, C_i > 0$ and $B > 0$, the unconstrained deadline problem becomes

$$\min_{\ell \leq \tau \leq u} \sum_i \left(\frac{A_i}{\tau_i} + C_i \tau_i\right) + B \max_{p \in \mathcal{P}} p^\top \tau. \tag{8}$$

Theorem 2 (Path-Flow Dual Bracket)**.** *For any dual flow vector $\rho \in B \operatorname{conv}(\mathcal{P})$, define the dual objective and duration response:*

$$D(\rho) = \sum_i \min_{\ell_i \leq t \leq u_i} \left[\frac{A_i}{t} + (C_i + \rho_i)t\right], \tag{9}$$

$$\tau_i(\rho) = \operatorname{clip}\left(\sqrt{\frac{A_i}{C_i + \rho_i}}, \ell_i, u_i\right). \tag{10}$$

Then strong duality holds: $\max_\rho D(\rho) = \min_\tau J(\tau)$. Moreover, every dual-feasible ρ provides an inspectable optimality gap:

$$0 \leq J(\tau(\rho)) - J^* \leq BM(\tau(\rho)) - \rho^\top \tau(\rho). \tag{11}$$

Proof. The set $\mathcal{P}$ contains every source-to-sink path incidence vector of the fixed DAG; $B \operatorname{conv}(\mathcal{P})$ consists exactly of nonnegative path mixtures whose total path weight is B. It is compact and convex. For $L(\tau, \rho) = \sum_i (A_i/\tau_i + C_i \tau_i) + \rho^\top \tau$, the duration box is also compact and convex, L is continuous and convex in τ, and it is affine in ρ. The minimax theorem interchanges min and max. Since $\max_{\rho \in B \operatorname{conv}(\mathcal{P})} \rho^\top \tau = BM(\tau)$, the primal is recovered. Coordinatewise minimization gives $D(\rho)$ and the clipped square-root response. For any feasible ρ, weak duality gives $D(\rho) \leq J^*$. At $\tau(\rho)$, $J(\tau(\rho)) - D(\rho) = BM(\tau(\rho)) - \rho^\top \tau(\rho)$. Combining the two inequalities yields the stated upper bound on primal suboptimality. □

Because $\nabla D(\rho) = \tau(\rho)$, a longest-path oracle on the DAG solves the linear subproblem in $O(m+r)$ time. A conditional gradient (Frank–Wolfe) update $\rho \leftarrow (1-\alpha)\rho + \alpha B p$

with bisection line search guarantees dual-feasible monotonic ascent.

V. Nonunitary Operational Certificates

We extend operational guarantees to open quantum systems governed by the Lindblad master equation:

$$\dot{\rho} = -\mathrm{i}[H(t), \rho] + \sum_{\mu} \left(L_\mu \rho L_\mu^\dagger - \tfrac{1}{2}\{L_\mu^\dagger L_\mu, \rho\} \right). \tag{12}$$

Let $\Delta H(t)$ denote coherent Hamiltonian mismatch, $N = \int \sum_\mu \|L_\mu(t)\|_\infty^2 dt$, and $C = \int \|\Delta H(t)\|_\infty dt$.

For clarity, F_{pur} below denotes the infimum, over normalized input states with an arbitrary untouched reference, of the squared overlap between the ideal output purification and the implemented output state.

Theorem 3 (Unitary-jump operational bound)**.** *Suppose coherent Hamiltonian mismatch vanishes and $L_\mu(t) = \sqrt{r_\mu(t)} V_\mu(t)$ with $V_\mu^\dagger V_\mu = I$. If $\Lambda = \int \sum_\mu r_\mu(t)\, dt \le R$, then*

$$F_{\mathrm{pur}} \ge e^{-R}, \tag{13}$$

$$\tfrac{1}{2}\|\mathcal{N} - \mathcal{U}\|_\diamond \le 1 - e^{-R}. \tag{14}$$

Proof. For every time, $\sum_\mu L_\mu^\dagger L_\mu = h(t) I$ with $h(t) = \sum_\mu r_\mu(t)$. The no-jump propagator is therefore $K_0 = e^{-\Lambda/2} U$, even when the unitary jumps do not commute with the control. The trajectory expansion is a sum of completely positive branches, and its no-jump branch is $e^{-\Lambda}\mathcal{U}$. As the complete map is trace preserving, the sum of all remaining branches is a completely positive map whose output trace is $(1 - e^{-\Lambda}) \operatorname{Tr} \rho$. After normalization it is a channel $\mathcal{Q}$ (the $\Lambda = 0$ case is immediate). Thus $\mathcal{N} = e^{-\Lambda}\mathcal{U} + (1 - e^{-\Lambda})\mathcal{Q}$. Tensoring with any reference preserves the decomposition. The output overlap with the ideal pure state is at least the weight of its ideal branch, proving $F_{\mathrm{pur}} \ge e^{-\Lambda} \ge e^{-R}$. Finally, $\|\mathcal{Q} - \mathcal{U}\|_\diamond \le 2$ implies the distance bound. □

Proposition 5 (General Markovian Operational Certificates)**.** *For general Markovian dynamics including nonunital amplitude damping (T_1 relaxation via $L = \sigma^-$, where $(\sigma^-)^\dagger \sigma^- \neq I$), with integrable bounds N and C, the implemented channel $\mathcal{N}$ and ideal channel $\mathcal{U}$ satisfy:*

$$\tfrac{1}{2}\|\mathcal{N} - \mathcal{U}\|_\diamond \le \min(1, N + C), \tag{15}$$

$$F_{\mathrm{pur}} \ge \max\left\{ (1 - N - C)_+, \left(1 - \tfrac{N}{2} - C\right)_+^2 \right\}. \tag{16}$$

Proof. Let $\Gamma(t) = \sum_\mu L_\mu(t)^\dagger L_\mu(t)$ and move to the ideal interaction picture, including an arbitrary untouched reference. The dissipator superoperator has diamond norm at most $2\sum_\mu \|L_\mu(t)\|_\infty^2$; the coherent commutator has diamond norm at most $2\|\Delta H(t)\|_\infty$. Duhamel's formula between CPTP propagators therefore gives $\|\mathcal{N} - \mathcal{U}\|_\diamond/2 \le N + C$, capped by one. For the second bound, the no-jump interaction-picture operator K_0 obeys $\dot{K}_0 = (-i\widetilde{\Delta H} - \widetilde{\Gamma}/2) K_0$. Its propagator is contractive, and integration gives $\|K_0 - I\|_\infty \le C + N/2$. On any normalized input purification $|\psi\rangle$, $|\langle\psi| K_0 \otimes I |\psi\rangle| \ge (1 - C - N/2)_+$. The no-jump contribution to the output overlap is the square of this amplitude, while all other Kraus contributions are nonnegative. The trace-distance bound independently yields an overlap of at least $(1 - N - C)_+$ for a pure ideal output. Taking the better floor and the infimum over inputs proves the claim. □

Our physical spin model uses $K_i = \theta_i P_i/2$. Always-on memory jumps are $\sqrt{1/T_{1q}}\sigma_q^-$ and $\sqrt{1/(2T_{\phi q})} Z_q$, while gate-dependent drive jumps are $\sqrt{\eta_i} K_i/\tau_i$ and $\sqrt{\gamma_i} P_i$. While Theorem 3 provides an exponential floor for unital jump generators, Proposition 5 rigorously bounds general nonunital T_1 relaxation.

VI. Discrete Pulse Libraries and Certificate Transfer

When hardware restricts controls to a discrete pulse library $\{d_{ik}\}_{k=1}^K \subset [\ell_i, u_i]$, duration selection is governed by binary indicators $x_{ik} \in \{0, 1\}$ with $\sum_k x_{ik} = 1$ and $\tau_i = \sum_k d_{ik} x_{ik}$.

For each pulse implementation $\mathcal{E}_{ik}$ and target $\mathcal{U}_i$, let J_{ik} be the unnormalized Choi matrix of their difference. Watrous's SDP [9] yields the dual upper bound $\delta_{ik} = \min\{1, \tfrac{1}{2}\|\operatorname{Tr}_{\mathrm{out}} |J_{ik}|\|_\infty\}$. Summing over disjoint gates with memory dissipation rate b_i gives compositional channel error $Q(x, M) = bM + \sum_{ik} (\delta_{ik} - b_i d_{ik}) x_{ik}$.

Proposition 6 (GP-to-Library Transfer Bound)**.** *Let D_{GP} be a continuous GP dual lower bound. Define residual errors $e_{ik} = (\delta_{ik} - b_i d_{ik} - a_i/d_{ik} - \gamma_i d_{ik})/R_0$. The discrete characterized optimum satisfies:*

$$J_{\mathrm{char}}^* \ge \max\left\{ 0, D_{\mathrm{GP}} + \sum_i \min_k e_{ik} \right\}. \tag{17}$$

Proof. Every library policy lies in the GP duration box. At such a policy, the characterized objective equals the GP objective plus $\sum_i e_{i,k(i)}$. Weak duality lower-bounds the former by D_{GP}, and each residual is bounded by its gatewise minimum. Minimizing preserves the inequality. □

VII. Scalable Observable-Constrained Scheduling

While dense superoperator simulation is limited to few qubits ($4^n \times 4^n$ dimensionality), quantum algorithms, stabilizer codes, and benchmarking sequences frequently target specific Pauli observables or stabilizer generators.

A. Clifford Adjoint Tracking on Stabilizer Circuits

Consider a Clifford circuit with stabilizer generators $\{S_k\}_{k=1}^n$. Under backward Heisenberg evolution through the circuit DAG, each generator transforms via symplectic updates. Restricting circuit gates to the Clifford group guarantees that each Pauli operator propagates deterministically to a Pauli operator at each time slice. At gate j with support $\mathrm{supp}(j)$, local Pauli errors propagate to the output. The sensitivity of observable S_k to noise at gate j is given by the

average anticommutation rate over non-identity local Paulis $\mathcal{P}_j^*$:

$$h_{kj} = \frac{1}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} [S_k(t_j) \text{ anticommutes with } P], \quad (18)$$

where $|\mathcal{P}_j^*| = 4^{|\operatorname{supp}(j)|} - 1$. In the declared model, an isotropic Pauli channel follows each ideal Clifford gate. Its formal semigroup generator is

$$\mathcal{L}_j(\rho) = \frac{\lambda_j(t)}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} (P\rho P - \rho), \quad (19)$$

distributing jump rates uniformly over all non-identity local Paulis $\mathcal{P}_j^*$. The quantity $\Lambda_j(\tau_j) = a_j/\tau_j + \gamma_j\tau_j$ is the **integrated jump intensity** of an isotropic Pauli channel applied after the ideal gate, rather than an ordinary discrete gate-error probability. The semigroup parameter need not represent simultaneous evolution under the physical drive Hamiltonian. For noise acting during a gate, the same scalar formula requires a commuting or otherwise Pauli-covariant interaction-picture generator; it is not established for general driven dynamics. In the Heisenberg picture, the adjoint generator acts on Pauli observable S_k as $\mathcal{L}_j^\dagger(S_k) = \frac{\lambda_j(t)}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} (PS_kP - S_k) = -2\lambda_j(t)h_{kj}S_k$, yielding $\dot{S}_k = -2\lambda_j(t)h_{kj}S_k$. Composing these post-gate channels and ideal Clifford gates yields the *exact within-model* scalar expectation formula:

$$\langle S_k \rangle = \exp\left(-2 \sum_j h_{kj}\Lambda_j(\tau_j)\right). \quad (20)$$

This exponential form is exact for the specified post-gate isotropic Pauli channels. The intensity is a parameter of the channel; the model does not prove that actual driven-gate noise is Pauli, isotropic, or confined to the end of a gate. Anisotropic Pauli noise would require component-wise rates and anticommutation tracking. Imposing $\langle S_k \rangle \geq \tau_k^{\text{target}}$ yields:

$$\sum_j h_{kj}\left(\frac{a_j}{\tau_j} + \gamma_j\tau_j\right) \leq -\frac{1}{2}\ln\tau_k^{\text{target}} \equiv \beta_k. \quad (21)$$

Because $h_{kj}, a_j, \gamma_j \geq 0$, (21) is an **exact posynomial constraint**!

B. Sub-Millisecond Analytic Infeasibility Filter

Before invoking any numerical solver, the gatewise unconstrained noise minimum is evaluated: $\tau_j^{\text{unc}} = \text{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ in $O(m)$ time. Given the precomputed sensitivity matrix $H \in \mathbb{R}^{K\times m}$, evaluating the minimum achievable noise vector $\mu = H(a/\tau^{\text{unc}} + \gamma\tau^{\text{unc}})$ requires a sparse matrix-vector multiplication taking $O(\text{nnz}(H))$ operations (at most $O(Km)$ across K target observables). If $\max_k(\mu_k - \beta_k) > 0$, the operational target is infeasible within the specified model in $O(\text{nnz}(H))$ time (with a median screening latency of 0.70 ms across all configurations, 2.44 ms at 128 qubits, and a recorded maximum of 5.57 ms), avoiding a numerical solve for instances rejected by the filter.

C. Certified Log-Grid Geometric Quantization

Hardware pulse controllers enforce discrete timing increments. For a geometric ratio $q > 1$ (e.g., $q = 1.02$), available durations form a grid $d_j^{(m)} = \ell_j q^m \leq u_j$. To guarantee that discrete quantization never violates the operational threshold, we solve the continuous posynomial program using an amplified exposure factor $\tilde{\gamma}_j = q\gamma_j$. Given continuous solution τ^*, discrete allocations are obtained by upward rounding with upper-bound box clamping:

$$d_j^* = \min\left(u_j, \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j)\rceil}\right). \quad (22)$$

Because $d_j^* \in [\tau_j^*, q\tau_j^*]$, the inverse drive term satisfies $a_j/d_j^* \leq a_j/\tau_j^*$, while decoherence satisfies $\gamma_j d_j^* \leq q\gamma_j\tau_j^* = \tilde{\gamma}_j\tau_j^*$. Consequently,

$$\sum_j h_{kj}\left(\frac{a_j}{d_j^*} + \gamma_j d_j^*\right) \leq \sum_j h_{kj}\left(\frac{a_j}{\tau_j^*} + \tilde{\gamma}_j\tau_j^*\right) \leq \beta_k, \quad (23)$$

guaranteeing that the quantized schedule satisfies the exact operational expectation $\langle S_k \rangle \geq \tau_k^{\text{target}}$ by mathematical construction when the strengthened constraints are solved with a verified numerical feasibility margin.

VIII. Experimental Results and Visualizations

A. Calibration-Snapshot Study and Feasibility

The saved *ibm_torino* snapshot supplies nominal gate lengths and average gate errors, not measured duration–error curves. We map each snapshot error to a total isotropic Pauli intensity at its nominal duration, then set the inverse-duration share by $\alpha \in \{0.2, 0.5, 0.8\}$; α is assumed, not fitted. The study includes 144 combinations of 4–128 qubits, graph and scrambled circuit families, two seeds, the three assumed shares, and absolute minimum stabilizer-expectation targets 0.95 and 0.99. All calculations are local. The model omits idle relaxation, crosstalk, leakage, and duration-dependent calibration data.

Table I separates 27 feasible solves from 117 analytically rejected targets. In this strict-target protocol, every 64- and 128-qubit instance is rejected before optimization. Thus those rows demonstrate pre-filter operation, not large-scale solver performance. The feasible subset ends at 32 qubits and has median GP solve time 69.86 ms. The 0.95/0.99 targets should not be interpreted as achievable hardware specifications for the deeper workloads. Numerical GP and direct convex formulations agree on feasible cases within the recorded tolerance, while the path-flow gap bounds apply to the continuous declared model. The up-to-50% reduction against nominal durations partly follows from the imposed lower box $\ell_i = 0.5\tau_i^{\text{base}}$; the stronger comparison is the optimal feasible uniform-duration policy.

TABLE I

OBSERVABLE-CONSTRAINED SCHEDULING ACROSS SCALES (IBM HERON ARCHITECTURE). FEASIBLE OPTIMIZATION RUNTIME IS REPORTED SEPARATELY FROM $O(\mathrm{nnz}(H))$ ANALYTIC SCREENING.

Qubits	Gates	Feas.	Infeas.	Feas. Solve (ms)	Screen (ms)	Dual Gap	$\lvert\Delta_{\mathrm{form}}\rvert$
4	14	9	15	11.8	0.18	4.27e-06	1.10e-09
8	31	6	18	19.5	0.59	1.15e-05	1.58e-09
16	64	6	18	281.8	0.87	1.81e-05	1.69e-09
32	131	6	18	624.4	1.33	2.15e-05	1.03e-09
64	269	0	24	N/A	1.29	N/A	N/A
128	545	0	24	N/A	2.44	N/A	N/A

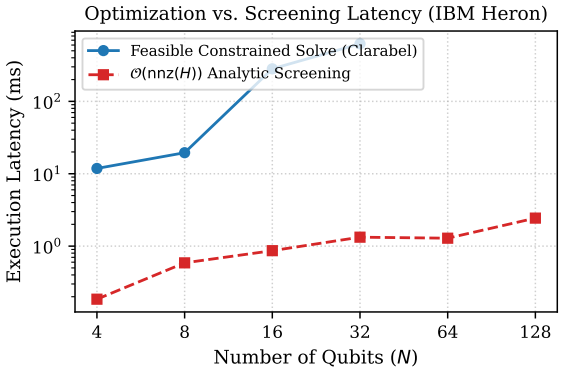


Fig. 1. Local solver times under a saved Heron calibration snapshot. The original absolute-target protocol has feasible solves only through 32 qubits; its 64- and 128-qubit timings measure analytic rejection, not optimization.

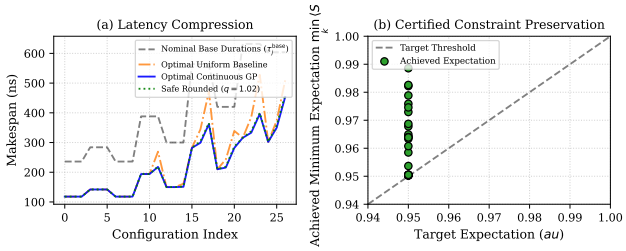


Fig. 2. Modeled performance using a saved Heron snapshot: (a) Latency compression showing makespan reduction across nominal calibrated base durations (τ_j^{base}) and the optimal uniform-duration baseline. (b) Certified preservation of operational expectation targets $\langle S_k \rangle \geq \tau$ after discrete log-grid rounding ($q = 1.02$).

B. Feasible Solves Through 128 Qubits

To measure solver behavior at larger sizes, we ran a separate, model-attainable protocol on 24 configurations at 32, 64, and 128 qubits (two families, two seeds, and assumed $\alpha \in \{0.5, 0.8\}$). For each instance, let e_ℓ be the minimum generator expectation at the lower-duration vector and $e_{\min}$ the minimum expectation at the coordinatewise minimum-hazard vector. We set the target to $\sqrt{e_\ell e_{\min}}$ and require $e_\ell < \sqrt{e_\ell e_{\min}} < e_{\min}$. This deliberately selects a nontrivial feasible target using the same model; it is a solver stress test, not external noise validation. We compare with the earliest *optimal uniform scaling* that satisfies that identical target. Targets vary by instance, so absolute fidelity levels are not comparable across sizes.

TABLE II

FEASIBLE MODEL-BASED SOLVES. TARGETS ARE CHOSEN SEPARATELY PER INSTANCE FROM ITS ASSUMED ATTAINABLE RANGE; TIMES ARE SINGLE-RUN WALL-CLOCK OBSERVATIONS, NOT HARDWARE TIMINGS.

Qubits	n	Gates	GP ms	Gain vs. uniform
32	8	64–200	1166.3	17.3%
64	8	132–412	2954.1	24.5%
128	8	266–830	6413.8	24.5%

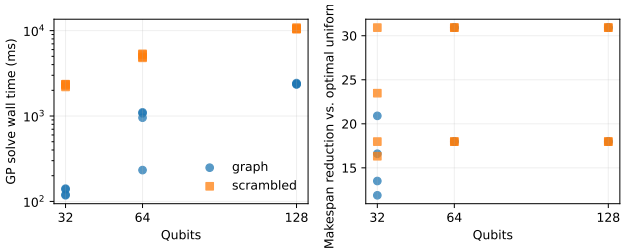


Fig. 3. Model-attainable 32–128-qubit continuous solves. Each point is one local solve; the right panel compares with an optimal uniform policy under the same instance-specific target. Neither panel measures device execution.

All 24 GP and direct-convex solves were feasible and their makespans agreed to a maximum relative difference below 3×10^{-7}. Table II reports descriptive medians of single-run wall time and modeled makespan reduction. The 128-qubit cases include up to 830 timed gates; their median observed solve time is 6.41 s. These runs address the earlier missing feasible large-scale solve, but do not establish linear scaling or physical speedup. The additional grid-rounding margin was not imposed in this study; a tightened margin may itself be infeasible or numerically difficult even when the original target is feasible.

C. Small-Channel Simulation and Matched Deadlines

Table III and Figure 4 compare finite pulse-library policies on the same simulated channels. The Lussi et al. scheduler is our adaptation of the published heuristic, not an independently supplied implementation. Unconstrained modeled process-fidelity changes in the balanced and high-drive settings are +3.97 and +15.34 percentage points, with corresponding latency increases of 32.6% and 217%. Under the fastest matched deadline, gains are small (reported group averages range from −0.009 to +0.026 percentage points), and some individual low-drive cases regress. The model and comparison therefore do not support a general fidelity

TABLE III

CHARACTERIZED MILP VERSUS ADAPTED PUBLISHED SLACK (48 CASES PER REGIME). FIDELITY CHANGES ARE PERCENTAGE POINTS (PP); INTERVALS ARE EXPLORATORY CLUSTER BOOTSTRAPS.

Regime	Mean pp	Interval	Regressions	Latency %
Low	-0.008	[-0.02,0.00]	16	0.05
Balanced	3.966	[3.10,4.86]	0	32.55
High	15.341	[9.88,20.67]	0	216.97

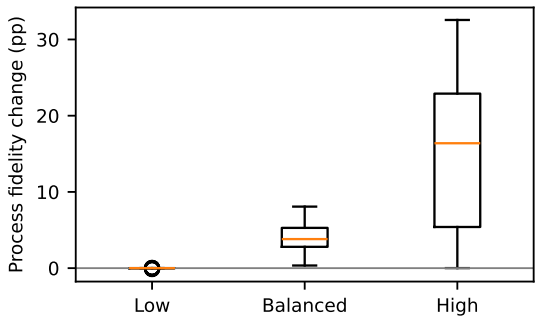


Fig. 4. Paired process-fidelity changes against adapted published slack scheduling [5] on the same simulated library, showing modeled fidelity and latency trade-offs; the matched-deadline result is reported in the text.

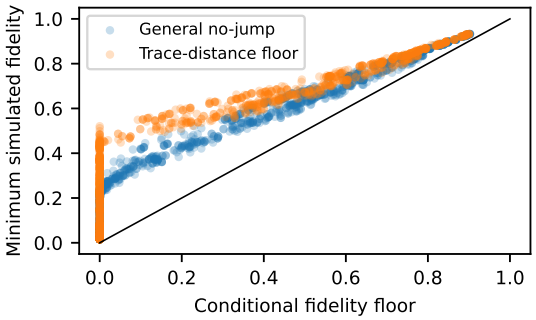


Fig. 5. General-jump certificate coverage. All simulated fidelities lie strictly above the analytical theoretical fidelity floors, checking implementation consistency with the assumed Lindblad model; this is not external physical validation.

TABLE IV

HELD-OUT MEASURED RB ERROR PER GATE, BY PULSE FAMILY. MSR IS MEAN SQUARED RESIDUAL DIVIDED BY REPORTED VARIANCE; SMALLER IS BETTER.

Pulse family	Fit MAE	Constant MAE	MSR
FAST DRAG-L	2.65×10^{-5}	3.47×10^{-5}	22.7
HD DRAG-L	7.76×10^{-5}	9.26×10^{-5}	40.6
Cosine DRAG-L	1.71×10^{-4}	1.84×10^{-4}	49.2
Cosine DRAG-P	3.71×10^{-4}	4.17×10^{-4}	82.5

advantage at fixed latency. The additional Qiskit scheduler studies below test both fixed-duration policies and idealized variable-duration GP policies in a common local evaluator.

D. Measured Error Versus Gate Duration

We tested the assumed inverse-plus-linear law on a publicly deposited experimental single-qubit randomized-benchmarking (RB) duration sweep by Hyyppä *et al.* [20]. Its Figure 3a archive provides measured mean error per gate, one-standard-deviation uncertainties, and durations for four calibrated pulse families. For each family, durations were sorted and alternating points allocated to fitting and held-out evaluation (54 points overall; 25 held out). Non-negative weighted least squares fitted $e(t) = A/t + Gt$ on the fit points using inverse reported variance. Table IV and Figure 6 report the held-out results. The law beats a constant-error predictor in mean absolute error for each family, but squared residuals measured in reported standard deviations are large. Thus its qualitative trade-off can describe these measurements while its two-parameter form does not adequately explain their reported uncertainty. Pulse-family-specific calibration and time-varying coherence are documented with the source data. These are external single-qubit measurements on a different platform, not a calibration or hardware validation of our three-qubit scheduler or IBM snapshot model.

An exploratory model comparison then considered $e_2(t) = A/t^2 + Gt$, motivated by the observed short-duration residuals. For each pulse family, we refitted nonnegative coefficients while leaving out one duration at a time. Table V shows lower mean absolute prediction error for e_2 than for $e_1(t) = A/t + Gt$ in all four families. However, the root mean squared residual divided by reported uncertainty remains well above one, and the inverse-square candidate was selected after inspecting the earlier holdout data. When fitted to the alternating training points, both laws selected the same minimum-predicted-error duration among held-out choices in every family. The predictive improvement therefore has no demonstrated decision benefit in this small dataset. These results generate a model hypothesis; they are not an independent confirmation or evidence of better circuit outcomes. Proposition 1 shows that testing a declared positive inverse power does not forfeit the continuous convex/GP structure.

a) Measured finite-library decision audit.: To test whether lower prediction error improves an actual choice, we fitted each law on the marked training durations and required it to select from the *same withheld measured duration options* within each pulse family. We also evaluated shape-preserving piecewise-cubic interpolation (PCHIP) as an exploratory empirical alternative. Table VI reports regret relative to the lowest reported RB error among withheld options. The two GP-compatible laws selected the same option in all four families. PCHIP selected the lowest observed option in three families and missed by 4.97×10^{-6} error per gate in HD DRAG-L; the inverse-power laws selected the lowest option in HD DRAG-L but incurred larger regret in FAST DRAG-L and Cosine DRAG-P. Thus a better *measured* duration choice is possible on this retrospective library, but it comes from an empirical finite-library predictor, not the proposed inverse-power GP law.

The reported one-sigma measurement error bars do not

TABLE V

EXPLORATORY LEAVE-ONE-DURATION-OUT ERROR-LAW COMPARISON. MAE COLUMNS ARE IN UNITS OF 10^{-4}; RMS IS THE ROOT MEAN SQUARED RESIDUAL DIVIDED BY SOURCE-REPORTED ONE-SIGMA UNCERTAINTY. THE INVERSE-SQUARE CANDIDATE WAS CHOSEN AFTER VIEWING EARLIER HOLDOUT DATA.

Pulse family	$1/t + t$ MAE	$1/t^2 + t$ MAE	$1/t^2 + t$ RMS
FAST DRAG-L	0.448	0.372	4.69
HD DRAG-L	1.345	1.153	14.74
Cosine DRAG-L	4.878	4.524	10.98
Cosine DRAG-P	6.309	5.977	8.58

TABLE VI

EXPLORATORY MEASURED FINITE-LIBRARY DURATION CHOICES. ENTRIES ARE MEASURED REGRET AGAINST THE LOWEST-ERROR WITHHELD OPTION, IN 10^{-5} RB ERROR PER GATE. THE EMPIRICAL INTERPOLATOR WAS EXAMINED AFTER EARLIER LAW RESULTS; THESE ARE NOT PROSPECTIVE GAINS.

Pulse family	$1/t + t$	$1/t^2 + t$	PCHIP
FAST DRAG-L	3.076	3.076	0.000
HD DRAG-L	0.000	0.000	0.497
Cosine DRAG-L	0.000	0.000	0.000
Cosine DRAG-P	3.452	3.452	0.000

cover model discrepancy. PCHIP's held-out root mean squared residual divided by the source sigma remains 1.92–6.82 across the four families. We estimated an additional per-family RMS discrepancy from leave-one-out predictions on *interior training points only*; the resulting diagnostic $1.96\sqrt{\sigma^2 + \tau^2}$ bands cover all withheld points in three families and five of six in Cosine DRAG-L. These are descriptive bands, not calibrated confidence or prediction intervals: there are few durations, adjacent errors may be correlated, and the empirical predictor was examined after the earlier law results. The test therefore does not establish prospective or target-device performance. A finite-library optimizer can use measured pulse options and conservative error inputs, but its discrete choice is outside the continuous GP optimality theorem.

E. Calibration Pairs and Idle-Noise Scheduling Stress Test

We extracted 566 reported gate-error/duration pairs for `sx`, `x`, and `cz` from the saved *ibm_torino* fake-backend calibration file. These are cross-sectional entries across gates and locations, sometimes with different parameter timestamps; they are not repeated measurements of one gate under controlled duration changes. Figure 7(a) displays the pairs descriptively. Fitting $a/t + \gamma t$ to this scatter would confound qubit, gate type, and calibration state, so no duration-response coefficients were inferred. A separate published two-qubit duration-sweep archive was identified [19], but its file service returned a storage-maintenance error at the time of this analysis; no values from it enter the results.

For a three-qubit native Clifford circuit, Qiskit's `ASAPScheduleAnalysis` and `ALAPScheduleAnalysis` were each followed by `PadDelay` or `PadDynamicalDecoupling` with two X pulses. All four schedules have the same 168-ns makespan and ideal output state. We independently propagated the density matrix under gate-local anisotropic Pauli errors, idle amplitude damping and dephasing, and, in one scenario, a Gaussian shot-static Z detuning acting during idles. Seven-node Gauss–Hermite quadrature per qubit integrated the static detuning. The gate error probabilities and detuning variance are declared stress-test assumptions, not fitted hardware parameters. Table VII gives ideal-state fidelities. Under Markovian Z-biased noise, DD adds faulty gates and lowers the ASAP fidelity; under shot-static detuning, ASAP+DD refocuses the long idle and improves it. ALAP's different idle placement also changes fidelity. These effects show that idle and noise anisotropy matter and that no scheduler dominates across the assumed noise regimes.

TABLE VII

MATCHED 168-NS QISKIT SCHEDULING BASELINES; SIMULATED IDEAL-STATE FIDELITY (NO HARDWARE JOBS).

Policy	Isotropic	Z-biased	Static Z + biased
ASAP	0.996804	0.995908	0.696027
ALAP	0.997087	0.996191	0.882678
ASAP+DD	0.996215	0.994426	0.994426
ALAP+DD	0.996281	0.995982	0.882493

F. Direct Equal-Deadline Comparison with Qiskit Scheduling

We next optimized the durations of the same five-gate, three-qubit circuit with the declared inverse-power law, using $p = 1$ or $p = 2$, a drive-error share $\alpha = 0.8$, duration bounds $[0.5, 1.5]$ times each nominal duration, and a 168-ns critical-path deadline. Nominal durations and errors came from the saved *ibm_torino* snapshot; the duration dependence and α are *assumed*, not calibrated. Each GP vector was placed ASAP or ALAP and padded to the common deadline. We compared it directly with Qiskit's ASAP/ALAP passes with either delay padding or two-X DD padding. All eight policies preserve the logical operation and meet 168 ns; the DD variants add two X gates. The GP variants require hypothetical alternate-duration native pulses, whereas Qiskit schedules the snapshot's fixed nominal gates. They therefore share the circuit, deadline, and evaluator, but not an experimentally available action set.

The evaluation independently propagates a density matrix through the explicit timed gate and delay list. A nominal gate error anchors each hypothetical gate-error curve $e(t) = e_0[\alpha(t_0/t)^p + (1-\alpha)t/t_0]$; a deliberately misspecified flat-error case tests sensitivity. Per-qubit Pauli errors are isotropic or Z biased; idle periods have $T_1 = 200\,\mu$s and $T_\phi = 250\,\mu$s, with optional shot-static Gaussian Z detuning of standard deviation 0.02 rad/ns. The static ensemble uses seven-node Gauss–Hermite quadrature per qubit. Coherent detuning and relaxation act during explicit delays only, and the model excludes driven-gate decoherence, crosstalk, leakage, and pulse-shape constraints. These are declared local simulation assumptions, not target-device predictions.

Table VIII reports deterministic simulated ideal-state fidelity. Under the aligned $p = 1$ Markovian model, GP-

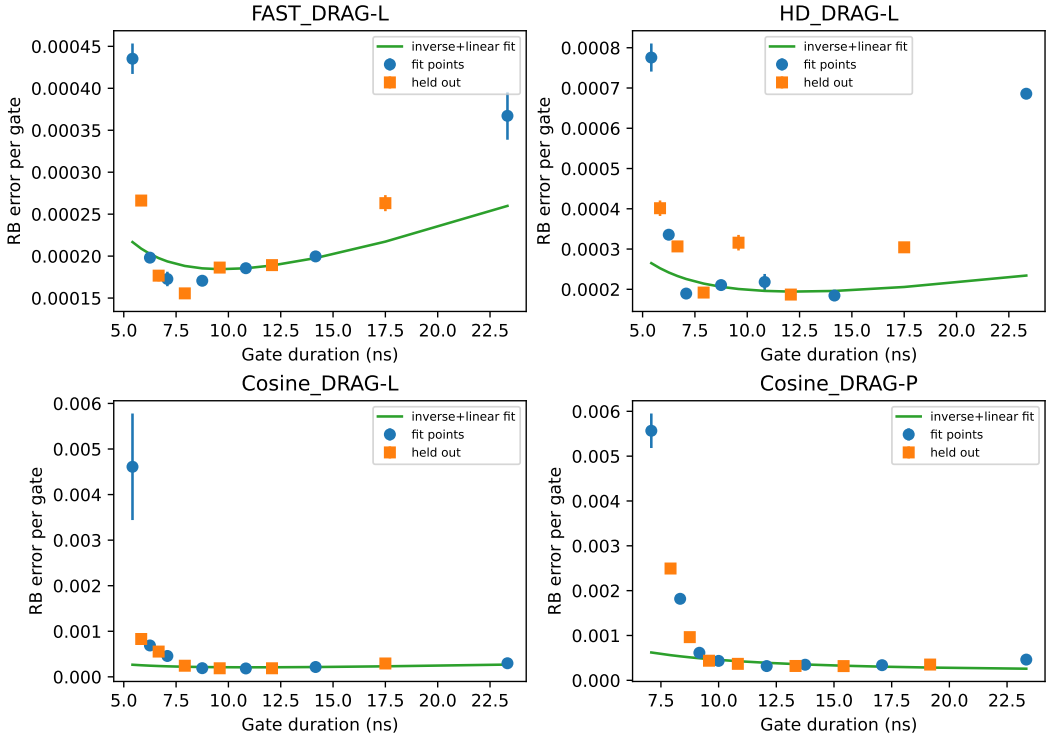


Fig. 6. Published experimental single-qubit RB gate error versus duration for four pulse families [20]. Circles fit the nonnegative inverse-plus-linear law; squares are held out. Error bars show source-reported one-standard-deviation uncertainties.

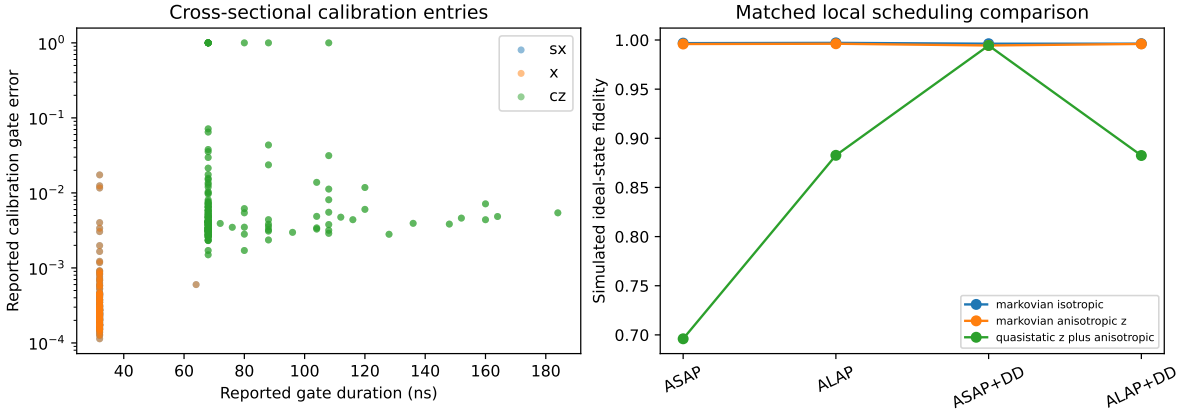


Fig. 7. Left: cross-sectional reported calibration pairs from a cached fake-backend file, not a controlled duration sweep. Right: matched-makespan local density-matrix simulation under three declared noise scenarios.

$p = 1$ ALAP reaches 0.993483 versus Qiskit ALAP's 0.993162, a small absolute gain. Under the $p = 2$ Z-biased Markovian model, GP-$p = 2$ ALAP reaches 0.994553 versus 0.994203. The flat-error case leaves only a tiny idle-placement difference. Under shot-static detuning, Qiskit ASAP+DD reaches 0.993448 while the best GP-only schedule reaches 0.938164. Seven- versus nine-node quadrature differs by at most 1.3×10^{-12} in these two leading static-noise policies. This rank reversal is a counterexample to general superiority of duration allocation without DD under an omitted idle process. The p=1 and p=2 GP vectors differ in this assumed model; it does not repair the unchanged held-out duration decisions in the published measurement sweep.

G. Exploratory Heron Idle-Noise Pilot

We executed a small, separate timing pilot on `ibm_marrakesh` through the free Open instance (job `db4ll9o4qg6s73c2buj0`). Four one-qubit circuits on physical qubit 0 were interleaved in two shuffled blocks, with 2,048 shots per circuit. Each prepared and measured an X-basis state with two logical Hadamards. The three Ramsey circuits inserted a 1, 20, or 80 μs idle delay; the fourth inserted two 40 μs delays separated by X, followed by a second X before the final Hadamard, preserving the ideal

TABLE VIII

DIRECT LOCAL COMPARISON AT A COMMON 168-NS DEADLINE: SIMULATED IDEAL-STATE FIDELITY UNDER DECLARED HYPOTHETICAL DURATION LAWS AND NOISE MODELS. NO HARDWARE JOBS OR MEASURED DURATION-RESPONSE CURVES.

Policy	$p = 1$, isotropic	$p = 2$, Z biased	Flat, Z biased	$p = 1$, static Z
GP $p = 1$ ASAP	0.993289	0.994205	0.994041	0.821760
GP $p = 1$ ALAP	0.993483	0.994399	0.994235	0.932671
GP $p = 2$ ASAP	0.993278	0.994357	0.994044	0.814316
GP $p = 2$ ALAP	0.993474	0.994553	0.994239	0.938164
Qiskit ASAP	0.992880	0.993920	0.993920	0.694797
Qiskit ALAP	0.993162	0.994203	0.994203	0.881003
Qiskit ASAP+DD	0.992584	0.993448	0.993448	0.993448
Qiskit ALAP+DD	0.992650	0.994042	0.994042	0.880861

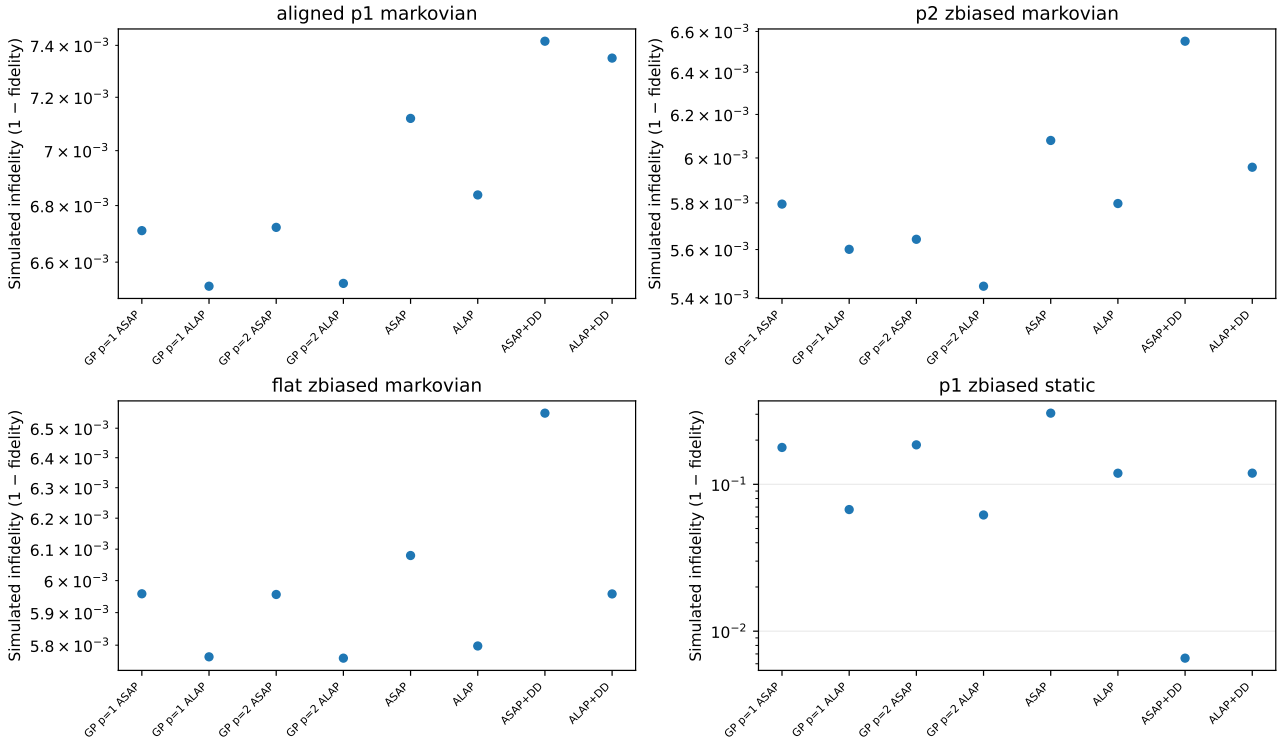


Fig. 8. Simulated infidelity ($1 - F$; logarithmic axis) for the eight policies in Table VIII. The horizontal positions are categorical. The static-detuning case exposes a large rank reversal favoring DD.

operation. Qiskit transpilation at optimization level zero retained all requested delays. The backend calibration snapshot immediately preceding the job reported $T_1 = 298.5$ μs and $T_2 = 40.4$ μs for that qubit. These are device-reported calibration values, not fitted parameters for our GP law.

Table IX gives the raw outcome fractions. Both blocks showed a higher zero frequency with the echo sequence than with the 80-μs Ramsey sequence; the blockwise differences were 0.1021 and 0.1201. This is exploratory evidence that idle dynamics and control placement matter on the target architecture. It is not a matched gate-count comparison, does not isolate a decay constant from coherent detuning and readout effects, and does not test a GP-selected gate duration. Shot-level exact-binomial intervals and the full counts are archived with the executable protocol; they exclude calibration drift. The job consumed 7 s of the free QPU allowance.

TABLE IX

EXPLORATORY *IBM_MARRAKESH* IDLE-NOISE PILOT ON PHYSICAL QUBIT 0: TWO BLOCKS PER CONDITION, 2,048 SHOTS PER BLOCK. $P(0)$ IS AN OBSERVED FREQUENCY, NOT CIRCUIT FIDELITY.

Circuit	Zero counts/shots	$P(0)$
Ramsey, 1 μs	4025/4096	0.9827
Ramsey, 20 μs	2655/4096	0.6482
Ramsey, 80 μs	1947/4096	0.4753
Echo, 80 μs	2402/4096	0.5864

H. Prespecified Heron Native-Gate Model Check

We next tested one narrower prerequisite for the duration study: whether the declared independent post-gate isotropic Pauli model, anchored to the backend's reported nominal gate errors and independent readout assignment errors, predicts outcomes for fixed native gates. Before submission, we saved a calibration snapshot and the predicted distributions for four small Clifford circuit families, compiled to physical

TABLE X

PRESPECIFIED NOMINAL NATIVE-GATE MODEL CHECK ON *IBM_MARRAKESH*; EACH FAMILY HAS THREE BLOCKS OF 2,048 SHOTS. RESIDUAL IS OBSERVED MINUS PRE-JOB PREDICTED EVENT PROBABILITY. THE TWO-PERCENTAGE-POINT TOLERANCE WAS DESCRIPTIVE, NOT A HYPOTHESIS TEST.

Family	Successes/6144	Predicted	Residual
Single return	6119	0.9685	+0.0274
Bell parity	6065	0.9619	+0.0253
Bell return	6000	0.9512	+0.0254
GHZ parity	5984	0.9201	+0.0538

TABLE XI

SECOND, HELD-OUT FIXED-NATIVE-GATE CHECK ON *IBM_MARRAKESH*: THREE 2,048-SHOT BLOCKS PER FAMILY. ENTRIES ARE TARGET-EVENT PROBABILITIES; CALIBRATED PREDICTIONS CHANGE ONLY MEASURED OPERATIONAL READOUT PARAMETERS. THE 0.02 TOLERANCE WAS DESCRIPTIVE.

Family	Observed	Snapshot	Calibrated
Single return	0.9937	0.9948	0.9951
Single-one return	0.9919	0.9917	0.9924
Bell parity	0.9849	0.9871	0.9872
Bell return	0.9740	0.9825	0.9851
GHZ parity	0.9723	0.9728	0.9755
GHZ return	0.9587	0.9632	0.9686

qubits 0–2 on `ibm_marrakesh`. Three shuffled blocks each contained one instance of every family, with 2,048 shots per instance (job `db4nov04qg6s73c2e8d0`). The prespecified events were return to `0`, Bell parity `00/11`, Bell return to `00`, and GHZ parity `000/111`. The prediction uses only nominal-duration gate errors, so it is independent of the unmeasured inverse-duration coefficient split.

All four discrepancies exceed the prespecified absolute tolerance of 0.02; the model underpredicts the observed success frequencies. The pre-job snapshot's qubit-0 probability of reporting 1 after preparing 0 was 0.03027, while the return circuit produced only 25 ones in 6,144 shots. This suggests a calibration-to-run mismatch or another unmodeled effect, but the experiment cannot distinguish calibration drift, readout treatment, and gate-noise misspecification. Shot-only Wilson intervals and blockwise frequencies are supplied with the raw counts and exact ISA circuits; neither accounts for calibration uncertainty. This result challenges quantitative transfer of the nominal model to this QPU. It does not test a GP duration choice or establish a hardware improvement.

We then froze a separate diagnostic before collecting new data: 18 readout-calibration circuits ($|0\rangle$ and $X|0\rangle$ on each of qubits 0–2, in three randomized blocks) and 18 held-out target circuits (six families in three randomized blocks), each at 2,048 shots. The calibration job `db4o3784qg6s73c2ekqg` completed before a SHA-256 lock of both the new backend-snapshot and empirically readout-adjusted predictions; only then was target job `db4o3bo4qg6s73c2ekvg` submitted. No target outcomes entered the fit. The $X|0\rangle$ control includes preparation error and is therefore an operational, not pure, readout estimate. Both Sampler twirling modes and dynamical decoupling were disabled. The gate model was unchanged.

All six readout-adjusted event residuals were below 0.02, but the snapshot prediction was more accurate overall (mean absolute event error 0.00286 versus 0.00476). The new backend calibration, time-stamped 02:47 local time, reported much smaller qubit-0 and qubit-2 readout errors than the earlier 02:28 snapshot (e.g., $P(1|0)$ on qubit 0 fell from 0.03027 to 0.00391). This change is consistent with calibration sensitivity, but does not prove why the first job disagreed. Three blocks and shot-only Wilson intervals do not quantify temporal drift. These fixed-gate observations neither validate the duration-error law nor demonstrate a GP schedule advantage.

TABLE XII

HELD-OUT HERON NATIVE-X IMPLEMENTATION COMPARISON AT 48 LOGICAL X GATES ON PHYSICAL QUBIT 1. ENTRIES ARE OBSERVED RETURN-TO-ZERO FREQUENCIES FROM THREE RANDOMIZED BLOCKS (6,144 SHOTS PER POLICY AND BASIS), NOT AVERAGE GATE FIDELITIES. EVERY POLICY HAS THE SAME NOMINAL DURATION.

Policy	X-basis return	Z-basis return
Slow X	0.9010	0.8296
Fast X, distributed delays	0.8826	0.8013
Fast X, tail delay	0.3052	0.9849
Fast X, two-X DD tail	0.8586	0.8753

I. Executable Native X-Duration Choice on Heron

The target exposes two implementations of the logical X gate: native `x` (36 ns) and `xslow` (1,000 ns) on physical qubits 0 and 1; IBM documents the latter as an extended-duration bit flip [18]. The backend reports the *same* nominal error for both, so those metadata do not characterize their duration dependence. We first submitted a prespecified, randomized characterization job (`db4ob04lf4us73c31490`): even sequences of 0, 8, 32, or 64 X gates, in X- and Z-basis return experiments on both qubits. Three blocks of 1,024 shots compared fast X, slow X, and fast X followed by a delay to match each slow gate's duration. At 32 gates on qubit 1 in the X basis, slow exceeded the equal-duration padded control by 0.0120 in all three blocks, but the shot-only 95% interval included zero. Fast X without duration matching remained better. We selected qubit 1 for a separately frozen holdout; this training-guided selection restricts generality.

The held-out job (`db4ocm2mb58s7389fnag`) used previously untested lengths of 16 and 48 X gates, both bases, and three randomized blocks of 2,048 shots. It compared slow X, distributed X-plus-delay padding, fast X followed by one tail delay, and the latter processed by Qiskit's ALAP analysis and two-X dynamical-decoupling pass. All four implementations had the same ideal unitary and checked nominal total duration; the two-X DD gate time was compensated in its tail delay. The compiled circuits and raw counts are archived. A protocol lock containing the training-data hash and shuffled target order was saved before QPU submission.

The frozen X-basis criterion was met: at 48 gates, slow X exceeded the equal-duration tail-delay baseline by 0.5959 and the standard two-X DD-tail baseline by 0.0425, with positive differences in all three blocks and shot-only 95%

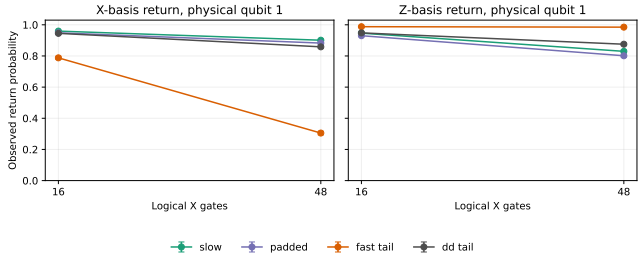


Fig. 9. Held-out physical X-duration comparison at equal nominal total durations. Error bars are shot-only 95% normal intervals and exclude device drift. The reversal between X- and Z-basis return makes a single scalar gate-error rate inadequate for these sequences.

intervals entirely above zero. Against distributed padding, the difference was 0.0184 (shot-only 95% interval 0.0074–0.0294). A paired three-block t interval for slow minus DD was 0.0288–0.0561, but three blocks do not establish temporal robustness. In the Z basis, slow X was worse than fast X with one tail delay and worse than DD. Thus the physical improvement is *observable- and schedule-specific*; it cannot be summarized as a universally more faithful X gate. The sharp basis-dependent rank reversal also challenges the isotropic, independent post-gate Pauli model used for the GP result. Because no GP optimizer chose the submitted circuit, this experiment supplies evidence of an implementable duration trade-off, not a GP hardware-performance demonstration.

J. Multi-Scale Statistical Verification

The following audits test implementation consistency against the declared model, rather than independent physical validity: For twelve deterministic fixed-DAG instances with three, five, or eight gates, inverse powers one or two, and optional deadlines, we additionally bounded each computed GP objective with a supporting-hyperplane linear program solved independently by HiGHS. The maximum relative upper–lower gap was 1.10×10^{-5}. This bounds numerical suboptimality for the specified convex objective in these small instances; it cannot validate its empirical gate-error coefficients or establish a hardware gain.

1) **Dense Master Equations ($n = 4$):** Comparison between analytical observable predictions and dense 16×16 density-matrix Lindblad simulation yielded a maximum deviation of 2.99×10^{-15}, confirming exact numerical agreement to machine precision.
2) **Forward Pauli-Frame Monte Carlo (16,384 shots):** Figure 10 displays forward Monte Carlo sampling deviations up to 32 qubits, verifying that empirical errors (≤ 0.0043) remain strictly inside simultaneous Hoeffding confidence bounds ($\epsilon \approx 0.033$–0.037 at $\alpha = 0.001$).

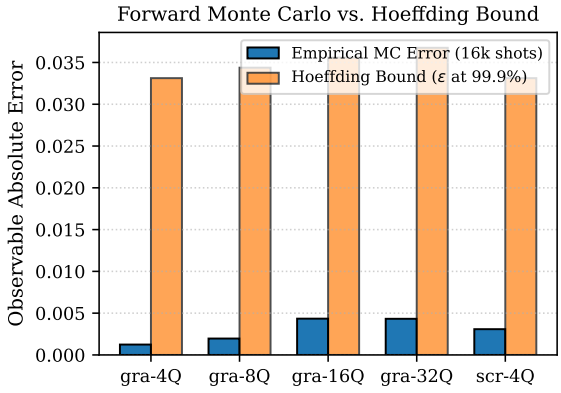


Fig. 10. Forward Pauli-frame Monte Carlo sampling validation (16,384 shots) across multi-qubit circuits. Empirical observable errors remain well below the simultaneous Hoeffding tolerance bounds (ϵ at 99.9% confidence).

K. Interpretation of the Comparative Evidence

The GP and ordinary convex formulations solve the same mathematical relaxation; their numerical agreement is a software and formulation check, not independent evidence that the physical model is accurate. The path-flow dual bracket bounds the optimum only for the declared fixed-DAG objective. The discrete-library transfer gap is sufficiently wide that it should not be read as a tight device-level performance certificate. Comparisons against the adapted slack heuristic are informative about modeled trade-offs, but unequal makespans confound unconstrained fidelity rankings. Accordingly, we treat matched-deadline results as the primary comparison and retain regressions. The 24 new feasible large-circuit instances establish that the continuous solver, rather than only its pre-filter, can run at 128 qubits under attainable modeled targets. They do not resolve coefficient identification, idle noise, or hardware control feasibility.

IX. Discussion and Compiler Integration

A. End-to-End Compiler Pass Architecture

We packaged the optimization engine as a native Qiskit compilation pass (`ConvexDurationAllocationPass`), subclassing `AnalysisPass`. The pass operates directly on the transpiler's `DAGCircuit` representation through a streamlined pipeline:

1) **DAG Extraction:** Extracts the directed acyclic dependency graph $\mathcal{D}$, identifying causal scheduling precedence between successive gates on overlapping qubit supports without changing the gate order.
2) **Analytic Infeasibility Pre-Screening:** In $O(\mathrm{nnz}(H))$ time, computes unconstrained gatewise noise minima $\tau_j^* = \mathrm{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ and evaluates $\max_k(\mu_k - \beta_k) \leq 0$ with precomputed sensitivity matrix H. If violated, it records an analytic infeasibility diagnostic without invoking the numerical solver.

3) **Convex Duration Allocation:** Formulates the posynomial program with safety exposure $q\gamma_j$ and invokes Clarabel/CVXPY, yielding a numerical continuous-duration solution on feasible instances.
4) **Log-Grid Quantization:** Maps continuous allocations τ^* to an abstract geometric pulse library via upward ceiling quantization with box clamping $d_j^* = \min(u_j, \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j)\rceil})$, with a conditional within-model feasibility bound.
5) **Property Set Annotation:** Annotates the pass manager's `property_set` with duration vectors, modeled makespan, and any computed dual gap; the prototype does not synthesize executable pulses.

B. Algorithmic Routine and Constrained Dual Certification

For observable-constrained duration allocation, let $w \in \mathbb{R}_+^K$ denote Lagrange multipliers for normalized observable constraints $\sum_j \frac{h_{kj}}{\beta_k}(a_j/\tau_j + q\gamma_j\tau_j) \le 1$. Grouping gate coefficients into $\tilde{A}_j(w) = a_j \sum_k \frac{w_k h_{kj}}{\beta_k}$ and $\tilde{C}_j(w) = q\gamma_j \sum_k \frac{w_k h_{kj}}{\beta_k}$, Lagrangian relaxation with dual DAG path pressure $\rho \in \frac{1}{M}\mathrm{conv}(\mathcal{P})$ yields the constrained dual lower bound:

$$D(w,\rho) = \sum_{j=1}^{m} \min_{\ell_j \le t \le u_j} \left[\frac{\tilde{A}_j(w)}{t} + (\tilde{C}_j(w) + \rho_j)t \right] - \sum_{k=1}^{K} w_k. \tag{24}$$

By weak duality, $D(w,\rho) \le M^*_{\mathrm{cont}}/\bar{M}$ for all dual-feasible (w,ρ), where $M^*_{\mathrm{cont}} = \mathrm{LongestPath}(\mathcal{D}, \tau^*)$ is the optimal makespan of the continuous relaxation. The path-flow Frank–Wolfe routine ascends on ρ to compute $D(w^*,\rho^*)$. The optimality gap strictly certifies convergence of this continuous solve:

$$\epsilon_{\mathrm{gap}} = \frac{M^*_{\mathrm{cont}}/\bar{M} - D(w^*,\rho^*)}{M^*_{\mathrm{cont}}/\bar{M}}. \tag{25}$$

Subsequent discrete log-grid rounding yields discrete makespan $M^*_{\mathrm{disc}} = \mathrm{LongestPath}(\mathcal{D}, d^*)$, with guaranteed upper bound $M^*_{\mathrm{disc}} \le qM^*_{\mathrm{cont}}$. The complete routine is formalized in Algorithm 1.

Algorithm 1: Observable-Constrained Gate Duration Allocation

Input: Circuit DAG $\mathcal{D} = (V, E)$, duration bounds $[\ell_j, u_j]$, noise rates a_j, γ_j, sensitivity matrix $H = (h_{kj}) \in \mathbb{R}^{K\times m}$, target observable fidelities τ_k^{target}, grid ratio $q > 1$, baseline makespan $\bar{M}$.
Output: Model-feasible discrete schedule $d^* \in \mathbb{R}^m$, discrete makespan M^*_{disc}, continuous relaxation makespan M^*_{cont}, relative optimality gap certificate ϵ_{gap}, or Infeasible diagnostic.

1) Set $\beta_k \leftarrow -\frac{1}{2}\ln \tau_k^{\mathrm{target}}$ for each target observable $k \in \{1,\dots,K\}$.
2) In $O(m)$ time, compute gatewise unconstrained minimizers $\tau_j^{\mathrm{unc}} \leftarrow \mathrm{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ and safety minimizers $\tau_j^{\mathrm{safe}} \leftarrow \mathrm{clip}(\sqrt{a_j/(q\gamma_j)}, \ell_j, u_j)$.
3) In $O(\mathrm{nnz}(H))$ time, compute lower-bound hazards: $\mu \leftarrow H(a/\tau^{\mathrm{unc}} + \gamma\tau^{\mathrm{unc}})$ and $\mu^{\mathrm{safe}} \leftarrow H(a/\tau^{\mathrm{safe}} + q\gamma\tau^{\mathrm{safe}})$.
4) **if** $\max_k(\mu_k - \beta_k) > 0$ **then return** Infeasible (target unachievable under baseline noise).
5) **if** $\max_k(\mu_k^{\mathrm{safe}} - \beta_k) > 0$ **then return** Infeasible (safety margin q uncertifiable).
6) Formulate convex posynomial program with safety exposure $\tilde{\gamma}_j \leftarrow q\gamma_j$:

$$\begin{aligned} \min_{\tau,f,M} \quad & M \\ \text{s.t.} \quad & \sum_j h_{kj}\left(\frac{a_j}{\tau_j} + q\gamma_j\tau_j\right) \le \beta_k, \quad \forall k, \\ & f_i + \tau_j \le f_j, \ \forall (i \to j) \in \mathcal{D}, \quad \tau_i \le f_i \le M, \\ & \ell_j \le \tau_j \le u_j, \quad \forall j. \end{aligned}$$

7) Solve via Clarabel/CVXPY and check the returned primal constraint residual; reject a numerical result that exceeds the implemented feasibility tolerance. **if** solver status is infeasible **then return** Infeasible (no schedule satisfies precedence and bounds under safety margin q).
8) Retrieve continuous optimum τ^* and dual multipliers $w^* \ge 0$.
9) Compute continuous makespan $M^*_{\mathrm{cont}} \leftarrow \mathrm{LongestPath}(\mathcal{D}, \tau^*)$.
10) Compute dual lower bound $D(w^*,\rho^*)$ via (24) and evaluate continuous gap:

$$\epsilon_{\mathrm{gap}} \leftarrow \frac{M^*_{\mathrm{cont}}/\bar{M} - D(w^*,\rho^*)}{M^*_{\mathrm{cont}}/\bar{M}}.$$

11) For each gate j, round upward with upper-bound box clamping:

$$d_j^* \leftarrow \min\left(u_j,\ \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j)\rceil}\right).$$

12) Compute model-based discrete makespan $M^*_{\mathrm{disc}} \leftarrow \mathrm{LongestPath}(\mathcal{D}, d^*)$.
13) **return** d^*, M^*_{disc}, M^*_{cont}, and primal–dual gap certificate ϵ_{gap}.

C. Scope, Missing Baselines, and Physical Validation

The geometric grid is an abstract library; it is not a measured set of alternate pulses on *ibm_torino*. A clock-aligned controller may require different rounding and an experimentally characterized noise response. The Qiskit implementation is an analysis pass that attaches duration results to a property set; it does not upload pulses or execute a modified schedule on hardware. The direct comparison uses Qiskit's published ASAP/ALAP and DD passes [17]; it holds circuit and deadline fixed, but GP has hypothetical continuous-duration actions unavailable to those passes. Its rank reversal under static detuning limits rather than establishes a superiority claim.

The largest source of uncertainty for the target device is the unmeasured split between inverse-duration and duration-proportional hazards. The snapshot determines one nominal point per gate, so multiple duration-error curves interpolate it equally well. Varying α probes this model uncertainty but cannot estimate α. The external measured single-qubit sweep shows that the assumed law can miss uncertainty-scale structure even where duration is varied experimentally; its coefficients cannot be transferred to IBM CZ gates. The Heron pilot varies *idle* duration, not driven-gate duration; it cannot identify this split either. The first fixed-native-gate QPU check fails its prespecified nominal-model tolerance in every family; the second passes a six-family descriptive tolerance after a backend calibration update. Readout recalibration from a separate job did not improve the second check. Thus the fixed-gate results are calibration-sensitive and cannot establish model transfer across times, much less the duration dependence needed for GP deployment. An inverse-square candidate improves exploratory prediction error while preserving convexity, but remains inconsistent with reported measurement uncertainty and has no independent confirmation. The retrospective PCHIP decision audit improves choices in two families relative to the GP laws, but its larger-than-reported residuals and post hoc selection rule out a prospective accuracy claim. The GP proof for the evaluated circuit study still assumes isotropic post-gate Pauli noise and excludes schedule-dependent idle processes. The separate simulator probes anisotropic errors, idle T_1/T_ϕ, and shot-static detuning during delays, but does not establish convexity for those additions. It omits detuning during driven gates, coherent crosstalk, leakage, and pulse-shape dependence. Dense Lindblad and Monte Carlo agreement only establish that separate implementations evaluate their shared assumed model consistently.

The native X experiment establishes that two supported gate durations can yield different physical returns under a matched schedule, and that a training-selected slow implementation can outperform equal-duration delayed-X and standard two-X DD controls on one held-out X-basis task. It does not show that the proposed inverse-plus-linear law chose the gate, that a GP solver improved hardware output, or that the ranking persists across bases: the Z-basis control reverses it. A gate-dependent, observable-aware model that accounts for driven evolution and coherent idle effects must be specified and fitted on one set of durations/circuits, then prospectively tested on independent choices and circuits. The existing two-point X library cannot identify a robust duration-response curve or test a continuous optimum. Wider hardware-compatible pulse libraries and matched, randomized comparisons against compiler baselines are necessary before asserting GP hardware benefit. The present contribution remains a reproducible convex timing method with conditional mathematical bounds plus a narrow physical gate-choice result.

X. Conclusion

For a fixed circuit DAG and declared inverse-plus-linear noise model, gate-duration allocation has an exact convex representation and an inspectable path-flow lower bound. The same convex structure accommodates a declared positive inverse-power model. Under post-gate isotropic Pauli noise on Clifford circuits, stabilizer-expectation targets produce convex constraints and conditional safe rounding for the evaluated law. The model-attainable study shows feasible continuous solves through 128 qubits, while the original strict-target study mainly demonstrates fast analytic rejection. Experimental duration sweeps reveal sizeable departures from the evaluated two-parameter law; an exploratory inverse-square variant reduces prediction error but does not explain reported uncertainty. A retrospective empirical finite-library predictor improves selected measured durations in two pulse families without validating the GP error law. The direct local GP-versus-Qiskit comparison shows only small modeled Markovian gains and a substantial regression against DD under shot-static detuning. The Heron pilot observes idle-time dependence and an echo response; two fixed-native-gate checks give inconsistent model-transfer outcomes across calibrations. A further training-selected, held-out native-X experiment demonstrates an X-basis return improvement over equal-duration delayed-X and standard two-X DD controls on one qubit, but a Z-basis reversal and the absence of a GP-selected physical schedule prevent attribution of that gain to the proposed optimizer. Broader measured duration libraries, a model that captures basis-dependent idle and driven noise, and prospective GP-versus-compiler hardware comparisons remain necessary for a GP performance claim.

Statements and Declarations

Funding. No funds, grants, or other support were received for this study.

Competing interests. The author declares no relevant financial or non-financial competing interests.

Data and code availability. The source, frozen protocols, raw QPU counts and QPY circuits, analysis scripts, benchmark outputs, and manuscript source are available at https://github.com/Youssef-Ahmed38/quantum-gp-circuit-optimization and in the accompanying supplementary archive (Online Resource 1).

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